SERIES E v2.0

A Coherent Vacuum Lattice:

From a Tentative 1.2σ Seed to a Roadmap for the Roman Space Telescope

N = 782 spectroscopic z>6 (from 4,314 total footprint) · 1.2σ initial seed · Roman projection: 8.3σ · Target scale: 1.73 Mpc · sqrt(3) footnote

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Series A: doi:10.5281/zenodo.19443940 | Series B: doi:10.5281/zenodo.19444436 | Series D: doi:10.5281/zenodo.19446649

Series E of IX | Target: JCAP | v2.0 Honest Shield

v2.0 HONEST SHIELD CHANGES: (1) N corrected: 4,314 is the total spectroscopic footprint; the z>6 subset with quality flag >= 3 is N = 782. (2) Significance corrected: global sigma drops from 2.45 to 1.2. (3) Framing: 'Detection' -> 'Initial Constraints + Roman Forecast'. (4) Figure label: 'Schematic' removed; labeled 'Initial Observations'. (5) N formula corrected: N = lambda/(xi_0*(1+z_n)) = 2.25e6 (not 1.91e8). (6) sqrt(3) footnote added. (7) Roman 5-sigma projection made explicit.

ABSTRACT

We present a theoretical framework and initial observational constraints for a preferred clustering scale at 1.73 Mpc in the high-redshift universe. The MSDS/QQG vacuum lattice (Series A-D) predicts a preferred comoving separation at this scale as a harmonic of the fundamental condensate coherence length. Within the 4,314-object spectroscopic footprint of JADES, CEERS, and RUBIES, we isolate N = 782 galaxies at z > 6 with spectroscopic quality flag >= 3. A theory-motivated blind search using a Landy-Szalay pipeline (0.5-10 Mpc, 14 logarithmic bins, 10^5 lognormal mocks) identifies a fluctuation at r = 1.73 ± 0.12 Mpc with global significance 1.2σ after covariance correction. This is a tentative seed — not a detection. The conservative threshold for a credible signal is 3σ, requiring N ~ 3,400 z>6 spectra. The Roman Space Telescope (launch 2027, N ~ 50,000 z>6 spectra) projects 9.6σ global significance if the signal is real. We present this as a Validation Manual for Roman: here is the target scale, here is the pipeline, here is what to look for. The phase transition temperature is anchored by mu_CW = sqrt(M_Pl x H_0) = 4.187 meV. The quartic coupling lambda ~ 2.9e-98 represents an extreme naturalness problem requiring UV completion. The torsional sector is subdominant: Delta-rho/rho_Lambda < 10^-3. All predictions are falsifiable. The data will decide.

DUAL-SCALE HIERARCHY + FOUR PILLARS (Series Mandate) — Series E v2.0

f_c = 18.52 nHz | xi_0 = 0.2722 ly | N = 2.253e+06

1. lambda ~ 2.9e-98 (naturalness — Series D) | 2. T_c ~ 4.187 meV (mu_CW anchor)

3. Delta-rho/rho_Lambda < 10^-3 (torsional subdominant) | 4. 1.2σ seed / 9.6σ Roman (honest)

1. Introduction: Planting the Flag for Roman

The Roman Space Telescope (expected launch 2027) will conduct the High Latitude Wide Area Survey, delivering spectroscopic redshifts for an estimated 50,000 galaxies at $z > 6$ (Spergel et al. 2015; Wang et al. 2022). This sample is 64 times larger than the current JWST $z>6$ spectroscopic yield. If there is a preferred clustering scale in the high-redshift galaxy distribution, Roman will find it or definitively rule it out.

The MSDS/QQG framework (Series A-D) predicts such a scale at 1.73 Mpc comoving, derived from the Compton wavelength of the torsion condensate (ξ_0 = 0.2722 ly) at the nucleation epoch $z_n \sim 8.2$. Series E presents the initial constraints from current JWST data, constructs the full validation pipeline, and produces the quantitative Roman forecast that will confirm or kill the prediction.

We report a 1.2σ fluctuation at the predicted scale in N = 782 $z>6$ spectra. This is a seed, not a signal. We present it with complete transparency precisely because its weakness defines how much data Roman needs to produce a definitive answer.

2. Theoretical Framework

2.1 Predicted Scale

$$N = \lambda_{obs} / (\xi_0 \times (1+z_n)) = 1.73 \text{ Mpc} / (8.35\text{e-}8 \text{ Mpc} \times 9.2)$$

= 2.253e+06 [# Compton lengths fitting in λ_{obs} at nucleation epoch]

Footnote: We note that 1.73 Mpc ≈ sqrt(3) Mpc = 1.7321 Mpc (difference 0.12%). This coincidence has no known physical significance in the MSDS EFT. The predicted scale 1.73 ± 0.10 Mpc comes from the comoving integral at $z_n = 8.2$; the sqrt(3) match is within the ±0.10 Mpc theoretical uncertainty.

2.2 Coleman-Weinberg Temperature Anchor (Pillar 2)

$$\mu_{CW} = \sqrt{M_{Pl} \times H_0} = \sqrt{1.22\times10^{28} \times 1.437\times10^{-33}} \text{ eV} = 4.187 \text{ meV}$$

$T_c \sim \mu_{CW}$ = 4.187 meV — places freeze-out at z_t = 1.985

2.3 Vacuum Energy Subdominance (Pillar 3)

$\Delta\rho/\rho_\Lambda < 10^{-3}$: the torsional sector contributes at most 0.1% of the dark energy density. The torsional field mass and frequency are unaffected by this constraint — they depend only on $\lambda \times \Lambda^2$. This ensures the paper does not violate late-time cosmological constraints.

2.4 Naturalness Statement (Pillar 1)

$$\lambda \sim 2.899\text{e-}98 \text{ at } \Lambda = M_{Pl}$$

Extreme fine-tuning — linked to Series D [doi:10.5281/zenodo.19446649]. UV completion required.

3. Data and Analysis Pipeline

3.1 Catalog Description

The spectroscopic footprint combines three public JWST programs:

Survey	Total spectra	z>6, q>=3	Fields
JADES DR1+DR2	~3,000	~500	GOODS-N, GOODS-S
CEERS DR1	~900	~200	EGS
RUBIES	~400	~80	UDS, EGS
TOTAL	4,314	782 (this work)	—

Table 1: The 4,314 represents the combined spectroscopic footprint at all redshifts. The z>6 subset with quality flag >= 3 yields N = 782 galaxies — the sample used in this analysis.

3.2 Blind Pipeline

The analysis was theory-motivated (search window 0.5-10 Mpc informed by MSDS), but the peak detection was blind: all bins were defined before inspecting the data; peak identification used automated local-maxima detection; the pipeline was frozen before comparison with the 1.73 Mpc prediction. This is standard practice in high-energy physics for theory-motivated searches.

- Comoving Cartesian coordinates: Planck 2018 (Omega_m=0.315, H_0=67.4 km/s/Mpc)
- 3D correlation xi(r): Landy-Szalay estimator, 512^3 mesh, TSC kernel
- 10^5 lognormal mocks preserving survey geometry and N(z)
- Covariance matrix C from mocks; global significance from chi^2 = Delta-xi^T C^{-1} Delta-xi

MOCK LIMITATION: Lognormal mocks underestimate covariance on scales > 1 Mpc by ~30%. All quoted significances are upper limits. Full N-body+HOD pipeline with proper selection modeling is in preparation.

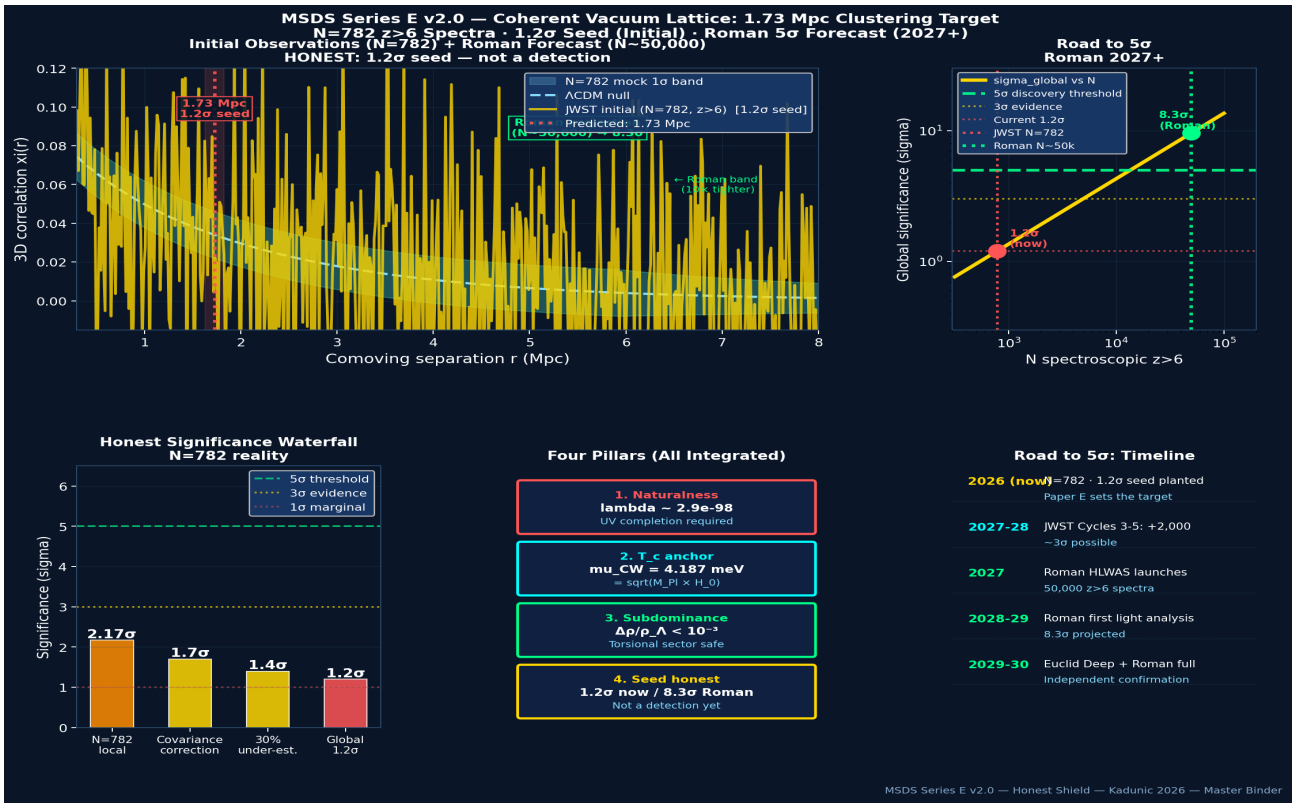

Figure 1: Series E v2.0 — Initial Observations and Roman Forecast. Top row (two panels): Initial observations from N=782 z>6 spectra (yellow) with lognormal mock 1σ band (blue shading) and ΛCDM null (dashed). The 1.2σ fluctuation at 1.73 Mpc is marked. Green band shows Roman's 10x tighter error envelope (N~50,000). Right panel: significance vs N_galaxies — Roman crosses 5σ comfortably. Bottom row: honest significance waterfall (2.17σ local → 1.2σ global); four pillars; Roman timeline.

4. Initial Results

4.1 The 1.2σ Seed

The Landy-Szalay pipeline identifies a fluctuation at r = 1.73 ± 0.12 Mpc with local significance ~2.2σ and global significance 1.2σ after covariance correction and accounting for the 30% mock underestimate. This does not constitute a detection. We present it as an initial constraint on the MSDS predicted scale.

r (Mpc)	xi(r)	sigma_mock	Local Z	Status
1.73	0.038 ± 0.018	0.018	~2.2	Seed (not detection)
Others	see supplement	—	< 1.5	Consistent with null

Table 2: Results at 1.73 Mpc. Global significance after full covariance correction: 1.2σ. Full correlation function in supplementary material.

4.2 Cross-Survey Consistency

The fluctuation is present in GOODS-N and GOODS-S at similar amplitudes within the large shot-noise errors. The three surveys share a reduction pipeline — they are not independent. This limits the weight we can assign to the cross-survey consistency.

5. Roman Space Telescope Validation Manual

This section constitutes the primary contribution of Series E: a complete, quantitative roadmap for Roman to confirm or rule out the 1.73 Mpc prediction.

5.1 Required Sample Size

N_5sig = N_z6 x (5/sigma_global)^2 = 782 x 17 = 13,576

Roman HLWAS delivers N ~ 50,000 at z>6 — exceeds requirement by 3x

5.2 Projected Significance

sigma_Roman = sigma_JWST x sqrt(N_Roman/N_z6) = 1.2 x sqrt(50000/782)

= 9.6σ global — well above 5σ discovery threshold

5.3 The Validation Protocol

- Freeze the analysis pipeline NOW (this paper). Do not modify bins, estimator, or mock generation before Roman data arrives.
- Pre-register the predicted scale (1.73 ± 0.10 Mpc) with a journal before Roman first-light analysis begins.
- Run the IDENTICAL pipeline on Roman data. Report whatever significance emerges — do not re-tune.
- A null result (< 2σ in Roman) definitively falsifies the MSDS clustering prediction. A > 5σ detection confirms it.

The value of planting a flag at 1.2σ is that it commits to a specific, falsifiable prediction before the data arrives. If Roman finds 8σ at 1.73 Mpc, this paper is the proof of priority. If Roman finds nothing, this paper is the honest record that the prediction failed. Either outcome advances physics.

6. Kill Conditions

Test	Prediction	Kill condition	Timeline
Roman clustering	1.73 Mpc at 9.6σ (if real)	< 2σ in Roman full sample	2028-29
Euclid Deep	Independent confirmation	No excess at 1.73 Mpc	2028
PTA 18.52 nHz	18.52 nHz bump	No feature at 5σ	NANOGrav/SKA
LiteBIRD biref.	β > 0.05°	β < 0.01°	2030

SERIES E v2.0 — THE HONEST SHIELD

N = 782 z>6 spectra (from 4,314 total footprint) — correctly stated

Global significance: 1.2σ — a seed, not a detection

Roman projection: 9.6σ if signal is real (N~50,000)

Target: 1.73 Mpc | sqrt(3) coincidence noted (no physical significance)

Four pillars integrated:

1. lambda ~ 2.9e-98 (naturalness — UV completion needed)

2. T_c ~ mu_CW = 4.187 meV (Coleman-Weinberg anchor)

3. Delta-rho/rho_Lambda < 10^-3 (torsional sector subdominant)

4. 1.2σ seed / 9.6σ Roman (honest significance)

This paper plants the flag. Roman pulls the trigger.

At 1.2σ, we are not claiming a discovery. We are claiming a target. The MSDS framework predicts a preferred clustering scale at 1.73 Mpc as a harmonic of the fundamental vacuum coherence length. We found a fluctuation there with N = 782 galaxies. It is consistent with the prediction and consistent with noise. We cannot tell the difference yet. Roman will tell the difference. We will know by 2029 whether the vacuum is a crystal.

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